SIMATS Online Programming Lab

Java Programming — CO4-AT2-Laboratory Performance
Kumaran S | Reg. No: 192372048RB

Q2. Sources of Events

Marks: 25 marks

Question: Write a program that demonstrates different sources of events (Button click, TextField input, Checkbox selection, Window closing). Display the event source and type.

Source Code

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import javax.imageio.ImageIO;
import java.io.File;

public class EventSourcesApp extends JFrame implements ActionListener, ItemListener, WindowListener {

    JButton clickButton;
    JTextField textField;
    JCheckBox checkBox;
    JTextArea logArea;

    public EventSourcesApp() {
        setTitle("Kumaran S (192372048RB) - Sources of Events");
        setSize(480, 380);
        setLayout(new BorderLayout());

        JLabel nameLabel = new JLabel("Kumaran S - 192372048RB", SwingConstants.CENTER);
        nameLabel.setFont(new Font("SansSerif", Font.BOLD, 16));
        nameLabel.setOpaque(true);
        nameLabel.setBackground(new Color(25, 55, 110));
        nameLabel.setForeground(Color.WHITE);
        nameLabel.setBorder(BorderFactory.createEmptyBorder(8, 8, 8, 8));
        add(nameLabel, BorderLayout.NORTH);

        JPanel controls = new JPanel(new FlowLayout());
        clickButton = new JButton("Click Me");
        textField = new JTextField(12);
        checkBox = new JCheckBox("Subscribe");

        controls.add(clickButton);
        controls.add(textField);
        controls.add(checkBox);
        add(controls, BorderLayout.CENTER);

        logArea = new JTextArea(10, 40);
        logArea.setEditable(false);
        add(new JScrollPane(logArea), BorderLayout.SOUTH);

        clickButton.addActionListener(this);
        textField.addActionListener(this);
        checkBox.addItemListener(this);
        addWindowListener(this);

        setDefaultCloseOperation(JFrame.DO_NOTHING_ON_CLOSE);
        setLocationRelativeTo(null);
        setVisible(true);
    }

    private void log(String msg) {
        logArea.append(msg + "\n");
    }

    public void actionPerformed(ActionEvent e) {
        Object src = e.getSource();
        if (src == clickButton) {
            log("Event Source: JButton (Click Me) | Event Type: ActionEvent (Button Click)");
```

```
        } else if (src == textField) {
            log("Event Source: JTextField | Event Type: ActionEvent (Text Input: \"" + textField.getText() +
        }
    }

    public void itemStateChanged(ItemEvent e) {
        String state = (e.getStateChange() == ItemEvent.SELECTED) ? "Selected" : "Deselected";
        log("Event Source: JCheckBox (Subscribe) | Event Type: ItemEvent (" + state + ")");
    }

    public void windowClosing(WindowEvent e) {
        log("Event Source: JFrame | Event Type: WindowEvent (Window Closing)");
        captureScreenshot("/home/claude/lab/screenshots/q2_output.png");
        System.exit(0);
    }
    public void windowOpened(WindowEvent e) {}
    public void windowClosed(WindowEvent e) {}
    public void windowIconified(WindowEvent e) {}
    public void windowDeiconified(WindowEvent e) {}
    public void windowActivated(WindowEvent e) {}
    public void windowDeactivated(WindowEvent e) {}

    private void captureScreenshot(String path) {
        try {
            Rectangle rect = this.getBounds();
            Robot robot = new Robot();
            Thread.sleep(300);
            java.awt.image.BufferedImage img = robot.createScreenCapture(rect);
            ImageIO.write(img, "png", new File(path));
        } catch (Exception e) {
            e.printStackTrace();
        }
    }

    public static void main(String[] args) throws Exception {
        EventSourcesApp app = new EventSourcesApp();
        SwingUtilities.invokeLater(() -> {
            app.clickButton.doClick();
            app.textField.setText("Hello Java");
            app.textField.postActionEvent();
            app.checkBox.setSelected(true);
        });
        Thread.sleep(700);
        app.dispatchEvent(new WindowEvent(app, WindowEvent.WINDOW_CLOSING));
        Thread.sleep(500);
    }
}
```

Output Screenshot